United States Patent Application Publication	20250275870
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	Wardle; John et al.

OCULAR IMPLANTS FOR DELIVERY INTO AN ANTERIOR CHAMBER OF THE EYE

Abstract

An ocular implant adapted to be disposed within Schlemm's canal of a human eye with a body extending along a curved longitudinal central axis in a curvature plane, a first strut on one side of the implant and a second strut on an opposite side of the implant, the circumferential extent of the first strut with respect to the plane of curvature being greater than the circumferential extent of the second strut with respect to the plane of curvature. The invention also includes methods of using the implant.

Inventors: **Wardle; John (San Clemente, CA), Schieber; Andrew T. (Tustin, CA)**

Applicant: **Alcon Inc.** (Fribourg, CH)

Family ID: **49380785**

Appl. No.: **19/211648**

Filed: **May 19, 2025**

Related U.S. Application Data

parent US continuation 18642324 20240422 parent-grant-document US 12336936 child US 19211648
parent US continuation 17314699 20210507 parent-grant-document US 11992437 child US 18642324
parent US continuation 15150175 20160509 parent-grant-document US 11026836 child US 17314699
parent US division 13793638 20130311 parent-grant-document US 9358156 child US 15150175
us-provisional-application US 61635104 20120418

Publication Classification

Int. Cl.: **A61F9/007** (20060101)

U.S. Cl.:

CPC **A61F9/00781** (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/642,324 filed Apr. 22, 2024, which is a continuation of U.S. application Ser. No. 17/314,699, filed May 7, 2021, now U.S. Pat. No. 11,992,437, which is a continuation of U.S. application Ser. No. 15/150,175, filed May 9, 2016, now U.S. Pat. No. 11,026,836, which is a division of U.S. application Ser. No. 13/793,638, filed Mar. 11, 2013, now U.S. Pat. No. 9,358,156, which application claims the benefit under 35 U.S.C. § 119 of U.S. Provisional Application No. 61/635,104, filed Apr. 18, 2012, the disclosures of which are incorporated by reference.

INCORPORATION BY REFERENCE
[0002] All publications and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference.
FIELD
[0003] The present invention relates generally to devices that are implanted within the eye. More particularly, the present invention relates to systems, devices and methods for delivering ocular implants into the eye.
BACKGROUND
[0004] According to a draft report by The National Eye Institute (NEI) at The United States National Institutes of Health (NIH), glaucoma is now the leading cause of irreversible blindness worldwide and the second leading cause of blindness, behind cataract, in the world. Thus, the NEI draft report concludes, “it is critical that significant emphasis and resources continue to be devoted to determining the pathophysiology and management of this disease.” Glaucoma researchers have found a strong correlation between high intraocular pressure and glaucoma. For this reason, eye care professionals routinely screen patients for glaucoma by measuring intraocular pressure using a device known as a tonometer. Many modem tonometers make this measurement by blowing a sudden puff of air against the outer surface of the eye.
[0005] The eye can be conceptualized as a ball filled with fluid. There are two types of fluid inside the eye. The cavity behind the lens is filled with a viscous fluid known as vitreous humor. The cavities in front of the lens are filled with a fluid know as aqueous humor. Whenever a person views an object, he or she is viewing that object through both the vitreous humor and the aqueous humor.
[0006] Whenever a person views an object, he or she is also viewing that object through the cornea and the lens of the eye. In order to be transparent, the cornea and the lens can include no blood vessels. Accordingly, no blood flows through the cornea and the lens to provide nutrition to these tissues and to remove wastes from these tissues. Instead, these functions are performed by the aqueous humor. A continuous flow of aqueous humor through the eye provides nutrition to portions of the eye (e.g., the cornea and the lens) that have no blood vessels. This flow of aqueous humor also removes waste from these tissues.
[0007] Aqueous humor is produced by an organ known as the ciliary body. The ciliary body includes epithelial cells that continuously secrete aqueous humor. In a healthy eye, a stream of

aqueous humor flows out of the anterior chamber of the eye through the trabecular meshwork and into Schlemm's canal as new aqueous humor is secreted by the epithelial cells of the ciliary body. This excess aqueous humor enters the venous blood stream from Schlemm's canal and is carried along with the venous blood leaving the eye.

[0008] When the natural drainage mechanisms of the eye stop functioning properly, the pressure inside the eye begins to rise. Researchers have theorized prolonged exposure to high intraocular pressure causes damage to the optic nerve that transmits sensory information from the eye to the brain. This damage to the optic nerve results in loss of peripheral vision. As glaucoma progresses, more and more of the visual field is lost until the patient is completely blind.

[0009] In addition to drug treatments, a variety of surgical treatments for glaucoma have been performed. For example, shunts were implanted to direct aqueous humor from the anterior chamber to the extraocular vein (Lee and Scheppens, "Aqueous-venous shunt and intraocular pressure," Investigative Ophthalmology (February 1966)). Other early glaucoma treatment implants led from the anterior chamber to a sub-conjunctiva! bleb (e.g., U.S. Pat. Nos. 4,968,296 and 5,180,362). Still others were shunts leading from the anterior chamber to a point just inside Schlemm's canal (Spiegel et al., "Schlemm's canal implant: a new method to lower intraocular pressure in patients with POAG?" Ophthalmic Surgery and Lasers (June 1999); U.S. Pat. Nos. 6,450,984; 6,450,984). More recent glaucoma treatment implants are designed to be advanced into and placed in Schlemm's canal. (See, e.g., U.S. Pat. No. 7,740,604; US 2011/0009958.)

SUMMARY OF THE DISCLOSURE

[0010] The present invention relates to methods and devices for treating glaucoma. In particular, the invention relates to an implant designed to extend from the anterior chamber of a human eye into Schlemm's canal and to support the tissue of Schlemm's canal to support flow of aqueous humor from the anterior chamber into Schlemm's canal to the outflow channels communicating with Schlemm's canal.

[0011] In one aspect, the invention provides an ocular implant adapted to be disposed within Schlemm's canal of a human eye and configured to support Schlemm's canal in an open state. The ocular implant has a body extending along a curved longitudinal central axis in a curvature plane, the body having a central channel bordered by an opening, first and second frames and a spine interposed between the first and second frames, the spine having a circumferential extent, the body having dimensions adapted to be fit within Schlemm's canal; each frame comprising a first strut on one side of the implant and a second strut on an opposite side of the implant, the first strut extending circumferentially beyond the circumferential extent of the spine on the one side of the implant and the second strut extending circumferentially beyond the circumferential extent of the spine on the other side of the implant, the circumferential extent of the first strut with respect to the plane of curvature being greater than the circumferential extent of the second strut with respect to the plane of curvature. In some embodiments, the body has a curved resting shape.

[0012] The implant may be adapted to bend preferentially in a preferential bending direction. In some embodiments, the preferential bending direction is in the curvature plane, and in some embodiments the preferential bending direction is not in the curvature plane.

[0013] In some embodiments, the circumferential extent of the first strut beyond the circumferential extent of the spine on the one side of the implant is greater than the circumferential extent of the second strut beyond the circumferential extent of the spine on the other side of the implant.

[0014] In embodiments in which the implant also has a third frame and a second spine interposed between the second and third frames, the second spine having a circumferential extent, the third frame having a first strut on one side of the implant and a second strut on an opposite side of the implant, the first strut and second strut of the third frame each having a circumferential extent greater than the circumferential extent of the second spine, the circumferential extent of the first strut with respect to the plane of curvature may be greater than the circumferential extent of the second strut with respect to the plane of curvature. The first, second and third frames may be substantially identical, and the first and second spines may be substantially identical. In some embodiments, the first spine is adapted to bend preferentially in a first bending direction, and the second spine is adapted to bend preferentially in a second bending direction different from the first bending direction.

[0015] In some embodiments, the plane of curvature intersects the spine. The spine may extend circumferentially in substantially equal amounts from the plane of curvature.

[0016] In some embodiments, the opening is an elongated opening extending longitudinally along the frames and the spine. The implant may also have a second opening bordered by the first and second struts of the first frame and a third opening bordered by the first and second struts of the second frame.

[0017] Another aspect of the invention provides a method of treating glaucoma in a human eye. The method may include the following steps: inserting a cannula through a cornea of the eye into an anterior chamber of the eye; placing a distal opening of the cannula in communication with Schlemm's canal of the eye; moving an ocular implant out of the cannula through the opening and into Schlemm's canal, the ocular implant having a central channel and first and second landing surfaces, the first and second landing surfaces being disposed on opposite sides of the central channel; and engaging a scleral wall of Schlemm's canal with the first and second landing surfaces such that reaction forces on the first and second landing surfaces from engagement with the scleral wall are substantially equal.

[0018] In some embodiments, the engaging step includes the step of engaging the scleral wall of Schlemm's canal with the first and second landing surfaces without substantially twisting the ocular implant.

[0019] In some embodiments, the implant has a resting shape forming a curve, and the method includes the step of orienting the implant curve with a curve of Schlemm's canal.

[0020] In some embodiments, the cannula is curved at a distal end, and the method includes the step of orienting the cannula for tangential delivery of the implant from the cannula into Schlemm's canal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. **1** is a stylized representation of an exemplary medical procedure in accordance with this detailed description.

[0022] FIG. **2** is an enlarged perspective view further illustrating delivery system **70** and eye **20** shown in the previous figure.

[0023] FIG. **3**A is a stylized perspective view illustrating the anatomy of an eye. FIG. **3**B is a stylized perspective view depicting the surface that defines the anterior chamber of the eye shown in FIG. **3**A.

[0024] FIG. **4** is a stylized perspective view showing Schlemm's canal and an iris of the eye shown in the previous figure.

[0025] FIG. **5**A is a photographic image showing a histology slide HS. Histology slide HS of FIG. **5**A was created by sectioning and staining tissue from a cadaveric eye. An ocular implant was implanted in Schlemm's canal of the cadaveric eye prior to sectioning. FIG. **5**B is a perspective view of the ocular implant used to generate the slide image shown in FIG. **5**A.

[0026] FIG. **6**A is a stylized line drawing illustrating histology slide HS shown in the previous figure. FIG. **6**B is a simplified cross-sectional view illustrating the eye from which the histology sample was taken. FIG. **6**A and FIG. **6**B are presented on a single page to illustrate the location of the histology sample relative to other portions of the eye.

[0027] FIG. **7**A is a stylized line drawing showing an ocular implant residing in a section of an eye including Schlemm's canal. FIG. **7**B is a section view showing the ocular implant prior to insertion into Schlemm's canal of the eye. FIG. **7**C is a perspective view showing the ocular implant being inserted into Schlemm's canal.
[0028] FIG. **8**A is a stylized line drawing showing an ocular implant according to the detailed description residing in a section of an eye including Schlemm's canal. FIG. **8**B is a section view showing the ocular implant of Figure A prior to insertion into Schlemm's canal of the eye. FIG. **8**C is a perspective view showing the ocular implant of FIGS. **8**A and **8**B being inserted into Schlemm's canal.
[0029] FIG. **9** is a perspective view showing the ocular implant of FIG. **7**.
[0030] FIG. **10** is an additional perspective view showing the ocular implant of FIG. **8**.
[0031] FIG. **11**A is a stylized perspective view showing a conical surface that is sized and positioned so as to intersect a hemispherical surface in two places. FIG. **11**B is a stylized perspective view showing an ocular implant disposed inside a chamber defined by a hemispherical surface.
[0032] FIG. **12**A, FIG. **12**B and FIG. **12**C are plan views of the ocular implant of FIG. **8** created using multiview projection.
[0033] FIG. **13**A is a plan view showing the ocular implant of FIG. **8**. FIG. **13**B is an enlarged section view taken along section line B-B shown in FIG. **13**A. FIG. **13**C is an additional enlarged section view taken along section line C-C shown in FIG. **13**A.
[0034] FIG. **14**A is a perspective view showing the ocular implant of FIG. **8**. A first plane and a second plane are shown intersecting the ocular implant in FIG. **14**A. FIG. **14**B is a plan view further illustrating the second plane shown in FIG. **14**A.
[0035] FIG. **15**A, FIG. **15**B and FIG. **15**C are multi-plan views of a yet another ocular implant in accordance with the detailed description.
[0036] FIG. **16**A is a plan view showing the ocular implant of FIG. **15**. FIG. **16**B is an enlarged section view taken along section line B-B shown in FIG. **16**A. FIG. **16**C is an additional enlarged section view taken along section line C-C shown in FIG. **16**A.
[0037] FIG. **17**A is a perspective view showing the ocular implant of FIG. **15**. A first, second and third planes are shown intersecting the ocular implant in FIG. **17**A. FIG. **17**B is a plan view further illustrating the second plane shown in FIG. **17**A. FIG. **1 7**C is a plan view further illustrating the third plane shown in FIG. **17**A.
[0038] FIG. **18**A is an additional perspective view of the ocular implant shown in the previous figure. The ocular implant **300** of FIG. **18**A includes a distal-most spine and a distal-most frame. In the exemplary embodiment of FIG. **18**A, the distal-most frame comprises a first strut and a second strut. FIG. **18**B is a stylized isometric view showing the profiles of the distal-most spine, the first strut and the second strut shown in FIG. **18**A.
[0039] FIG. **19**A and FIG. **19**B are perspective views showing distal portions of the ocular implant of FIG. **8** and the ocular implant of FIG. **15**, respectively.
[0040] FIG. **20** is a perspective view showing yet another exemplary ocular implant in accordance with the detailed description.
[0041] FIG. **21** is a stylized perspective view showing Schlemm's canal encircling an iris. For purposes of illustration, a window is cut through a first major side of Schlemm's canal in FIG. **21**. Through the window, an ocular implant can be seen residing in a lumen defined by Schlemm's canal.
[0042] FIG. **22** is an enlarged cross-sectional view further illustrating Schlemm's canal SC shown in FIG. **21**.
DETAILED DESCRIPTION
[0043] The following detailed description should be read with reference to the drawings in which similar elements in different drawings are numbered the same. The drawings, which are not

necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the invention.

[0044] FIG. **1** is a stylized representation of an exemplary medical procedure in accordance with this detailed description. In the exemplary procedure of FIG. **1**, a physician is treating an eye **20** of a patient P. In the exemplary procedure of FIG. **1**, the physician is holding a hand piece of a delivery system **70** in his or her right hand RH. The physician's left hand LH is holding the handle H of a gonio lens **23** in the exemplary procedure of FIG. **1**. It will be appreciated that some physicians may prefer holding the delivery system hand piece in the right hand and the gonio lens handle in the left hand.

[0045] During the exemplary procedure illustrated in FIG. **1**, the physician may view the interior of the anterior chamber using gonio lens **23** and a microscope **25**. Detail A of FIG. **1** is a stylized simulation of the image viewed by the physician. A distal portion of a cannula **72** is visible in Detail A. A shadow-like line indicates the location of Schlemm's canal SC which is lying under various tissue (e.g., the trabecular meshwork) that surround the anterior chamber. A distal opening of cannula **72** is positioned near Schlemm's canal SC of eye **20**.

[0046] Methods in accordance with this detailed description may include the step of advancing the distal end of cannula **72** through the cornea of eye **20** so that a distal portion of cannula **72** is disposed in the anterior chamber of the eye. Cannula **72** may then be used to access Schlemm's canal of the eye, for example, by piercing the wall of Schlemm's canal with the distal end of cannula **72**. A distal opening of cannula **72** may be placed in fluid communication with a lumen defined by Schlemm's canal. The ocular implant may be advanced out of the cannula and into Schlemm's canal. Insertion of the ocular implant into Schlemm's canal may facilitate the flow of aqueous humor out of the anterior chamber of the eye.

[0047] FIG. **2** is an enlarged perspective view further illustrating delivery system **70** and eye **20** shown in the previous figure. In FIG. **2**, cannula **72** of delivery system **70** is shown extending through a dome-shaped wall **90** of eye **20**. The dome shaped wall includes the cornea **36** of eye **20** and scleral tissue that meets the cornea at a limbus of the eye. A distal portion of cannula **72** is disposed inside the anterior chamber AC defined by the dome-shaped wall **90**. In the embodiment of FIG. **2**, cannula **72** is configured so that a distal opening of cannula **72** can be placed in fluid communication with Schlemm's canal. In the embodiment of FIG. **2**, the distal end of cannula **72** is curved so that the distal opening of the cannula can be inserted at least partially into Schlemm's canal along a tangential approach.

[0048] In the embodiment of FIG. **2**, an ocular implant is disposed in a passageway defined by cannula **72**. Delivery system **70** includes a mechanism that is capable of advancing and retracting the ocular implant along the length of cannula **72**. The ocular implant may be placed in Schlemm's canal of eye **20** by advancing the ocular implant through the distal opening of cannula **72** while the distal opening is in fluid communication with Schlemm's canal.

[0049] FIG. **3**A is a stylized perspective view illustrating a portion of eye **20** discussed above. Eye **20** includes an iris **30** defining a pupil **32**. In FIG. **3**A, eye **20** is illustrated in a cross-sectional view created by a cutting plane passing through the center of pupil **32**. Eye **20** includes a dome-shaped wall **90** having a surface **92** defining an anterior chamber AC. In FIG. **3**A, surface **92** is shown having a generally hemispherical shape. Dome-shaped wall **90** of eye **20** comprises a cornea **36** and scleral tissue **34**. The scleral tissue **34** meets the cornea **36** at a limbus **38** of eye **20**. Additional scleral tissue **34** of eye **20** surrounds a posterior chamber PC filled with a viscous fluid known as vitreous humor. A lens **40** of eye **20** is located between anterior chamber AC and posterior chamber PC. Lens **40** is held in place by a number of ciliary zonules **42**.

[0050] Whenever a person views an object, he or she is viewing that object through the cornea, the aqueous humor, and the lens of the eye. In order to be transparent, the cornea and the lens can include no blood vessels. Accordingly, no blood flows through the cornea and the lens to provide nutrition to these tissues and to remove wastes from these tissues. Instead, these functions are performed by the aqueous humor. A continuous flow of aqueous humor through the eye provides nutrition to portions of the eye (e.g., the cornea and the lens) that have no blood vessels. This flow of aqueous humor also removes waste from these tissues.

[0051] Aqueous humor is produced by an organ known as the ciliary body. The ciliary body includes epithelial cells that continuously secrete aqueous humor. In a healthy eye, a stream of aqueous humor flows out of the eye as new aqueous humor is secreted by the epithelial cells of the ciliary body. This excess aqueous humor enters the blood stream and is carried away by venous blood leaving the eye.

[0052] With reference to FIG. **3**A, it will be appreciated that Schlemm's canal SC is disposed inside anterior chamber AC. Schlemm's canal SC is a tube-like structure that encircles iris **30**. In the illustration of FIG. **3**A, the cutting plane passing through the center of pupil **32** has also passed through Schlemm's canal. Accordingly, two laterally cut ends of Schlemm's canal SC are visible in the cross-sectional view of FIG. **3**A. In a healthy eye, aqueous humor flows out of anterior chamber AC and into Schlemm's canal SC. Aqueous humor exits Schlemm's canal SC and flows into a number of collector channels. After leaving Schlemm's canal SC, aqueous humor is absorbed into the venous blood stream and carried out of the eye.

[0053] Because of the position of Schlemm's canal SC within the anterior chamber AC, a Schlemm's canal access cannula inserted through the cornea **36** and anterior chamber AC is likely to approach the plane of Schlemm's canal at an approach angle that is greater than zero. Thus, for example, when using a curved cannula (such as the one shown in FIG. **2**) to deliver an implant into Schlemm's canal, the plane of curvature of the cannula will form a non-zero angle with the plane of Schlemm's canal.

[0054] FIG. **3**B is a stylized perspective view depicting the surface **92** that defines anterior chamber AC of the eye shown in FIG. **3**A. In FIG. **3**B, surface **92** is shown having a generally hemispherical shape. FIG. **3**B may be used to illustrate some fundamental geometric concepts that will be used below to describe the various ocular implant structures. Geometry is a branch of mathematics concerned with the properties of space and the shape, size, and relative position of objects within that space. In geometry, a sphere is a round object in three-dimensional space. All points on the surface of a sphere are located the same distance r from a center point so that the sphere is completely symmetrical about the center point. In geometry, a point represents an exact location. A point is a zero-dimensional entity (i.e., it has no length, area, or volume). Geometrically speaking, at any point on a spherical surface, one can find a normal direction which is at right angles to the surface. For a spherical surface all normal directions intersect the center point of the sphere. Each normal direction will also be perpendicular to a line that is tangent to the spherical surface. In FIG. **3**B, a normal line N is illustrated using dashed lines. Normal line N is at right angles to spherical surface **92**. Normal line N is also perpendicular to a reference line TAN. Reference line TAN is tangent to spherical surface **92** in FIG. **3**B.

[0055] As shown in the previous figure, Schlemm's canal is disposed inside anterior chamber AC. An exemplary method in accordance with this detailed description may include the step of advancing a distal portion of a cannula into the anterior chamber of the eye. The cannula may then be used to access Schlemm's canal, for example, by piercing the wall of Schlemm's canal with the distal end of the cannula. An ocular implant may be advanced out of the distal opening of the cannula and into Schlemm's canal. An exemplary path **94** taken by an ocular implant as it follows Schlemm's canal along surface **92** is illustrated using a row of dots in FIG. **3**B.

[0056] As the ocular implant advances into Schlemm's canal, the ocular implant may press against the outer major wall of Schlemm's canal and the dome-shaped wall that defines the anterior chamber of the eye. As the body of the ocular implant presses against the dome-shaped wall of the eye, the dome-shaped wall provides support for Schlemm's canal and the ocular implant. The support provided by the dome-shaped wall may be represented by force vectors. The direction of these force vectors may be at right angles to points on the spherical surface that defines the anterior chamber. The dome shaped wall comprises scleral tissue that is firmer than the tissue of Schlemm's canal wall. Accordingly, the outer major wall of Schlemm's canal may be supported by the dome shaped wall as the ocular implant advances into Schlemm's canal.

[0057] During delivery, it is desirable that the ocular implant follow the lumen of Schlemm's canal as it is advanced out the distal opening of the cannula. The ability of the ocular implant to be advanced into and follow the lumen of Schlemm's canal may be referred to as trackability. Characteristics of an ocular implant that affect trackability include axial pushability, lateral flexibility, and overall shape with respect to the shape of Schlemm's canal (e.g., radius of curvature and cross-section profile). Axial pushability generally concerns the ability of an ocular implant to transmit to the distal end of the ocular implant an axial force applied to the proximal end of the ocular implant. Lateral flexibility concerns the ease with which the ocular implant body can bend to conform to the shape of the lumen. Trackability may be adversely effected when twisting forces are applied to a curved body. For example, twisting the body of a curved ocular implant about its longitudinal axis may cause the curved body to steer away from a desired path.

[0058] FIG. **4** is a stylized perspective view further illustrating Schlemm's canal SC and iris **30** shown in FIG. **3**A. Schlemm's canal SC and iris **30** are disposed inside the anterior chamber AC of the eye. The surface **92** that defines the anterior chamber AC of eye **20** is depicted using dashed lines in FIG. **4**. In the exemplary embodiment of FIG. **4**, Schlemm's canal SC and iris **30** are shown in cross-section, with a cutting plane passing through the center of a pupil **32** defined by iris **30**. Schlemm's canal SC comprises a first major side **50**, a second major side **52**, a first minor side **54**, and a second minor side **56**. Schlemm's canal SC forms a ring around iris **30** with pupil **32** disposed in the center of that ring. With reference to FIG. **4**, it will be appreciated that first major side **50** is on the outside of the ring formed by Schlemm's canal SC and second major side **52** is on the inside of the ring formed by Schlemm's canal SC. Accordingly, first major side **50** may be referred to as an outer major side of Schlemm's canal SC and second major side **52** may be referred to as an inner major side of Schlemm's canal SC. With reference to FIG. **4**, it will be appreciated that first major side **50** is further from pupil **32** than second major side **52**.

[0059] FIG. **22** is an enlarged cross-sectional view further illustrating Schlemm's canal SC. Schlemm's canal SC includes a wall W defining a lumen **58**. The shape of Schlemm's canal SC is somewhat irregular, and can vary from patient to patient. The shape of Schlemm's canal SC may be conceptualized as a cylindrical-tube that has been partially flattened. The cross-sectional shape of lumen **58** may be compared to the shape of an ellipse. A major axis **60** and a minor axis **62** of lumen **58** are illustrated with dashed lines in FIG. **22**.

[0060] The length of major axis **60** and minor axis **62** can vary from patient to patient. The length of minor axis **62** is between one and thirty micrometers in most patients. The length of major axis **60** is between one hundred and fifty micrometers and three hundred and fifty micrometers in most patients.

[0061] With reference to FIG. **22**, Schlemm's canal SC has a first major side **50**, a second major side **52**, a first minor side **54**, and a second minor side **56**. In the exemplary embodiment of FIG. **22**, first major side **50** is longer than both first minor side **54** and second minor side **56**. Also in the exemplary embodiment of FIG. **22**, second major side **52** is longer than both first minor side **54** and second minor side **56**.

[0062] An exemplary path **94** taken by an ocular implant as it follows Schlemm's canal along surface **92** is illustrated using a row of dots in FIG. **4**. As the ocular implant advances into Schlemm's canal, the ocular implant may press against the outer major wall of Schlemm's canal and the dome-shaped wall that defines the anterior chamber. More particularly, one or more surfaces (e.g., on the struts) of the ocular implant may press against surface **92**, i.e., scleral tissue forming part of the dome-shaped wall of the eye, as the implant moves into and along Schlemm's canal. The scleral tissue provides support for Schlemm's canal and the ocular implant as it is advanced into and along Schlemm's canal. The support provided by the scleral tissue may be represented by one or more force vectors with each force vector being at right angles to a point on the spherical surface that defines the anterior chamber of the eye; these force vectors act on the implant to balance the forces generated by the implant as the implant is inserted into and advanced along Schlemm's canal.

[0063] The interaction between the implant's structure and the tissue forming Schlemm's canal can affect how the implant behaves as it is inserted into and advanced along Schlemm's canal. For example, the implant may have surfaces (hereinafter, “landing surfaces”) that engage scleral tissue within Schlemm's canal as the implant is inserted into and advanced along Schlemm's canal. If the force vector on a landing surface on one side of the implant exceeds the force vector on an opposite side of the implant, the implant may bend or twist as it is advanced. In addition, if the implant has a preset curve, any bending or twisting of the implant may direct the curve away from, instead of along, the curve of Schlemm's canal. Also, the implant may have a preferential bending plane that will affect the orientation of the implant within a curved insertion cannula and with the curve of Schlemm's canal as well as the implant's response to force vectors on its landing surfaces.

[0064] For example, an ocular implant in accordance with the present detailed description may include a plurality of spines and a plurality of landing surfaces that seat against the inner surface of the dome shaped wall that encloses the anterior chamber so that the dome shaped wall provides supporting normal forces to the landing surfaces. The ocular implant may be configured such that a net twisting moment applied to each spine by the normal forces supporting the landing surfaces during implantation is reduced or is substantially zero. The ocular implant may also be configured such that the normal forces supporting the landing surfaces primarily or exclusively act to guide each spine along the preferential bending plane thereof.

[0065] FIG. **5**A is a photographic image showing a histology slide HS. Histology slide HS of FIG. **5** was created by sectioning and staining tissue sampled from a cadaveric eye. An ocular implant **500** was implanted in Schlemm's canal SC of the cadaveric eye prior to sectioning. The photograph of FIG. **5**A was created while examining the section of tissue using a light microscope. FIG. **5**B is a drawing of the ocular implant **500** used in FIG. **5**A. Similar to the ocular implants described in, e.g., U.S. Pat. No. 7,740,604, US Publ. No. 2009/0082860, U.S. Pat. No. 8,372,026 and US Publ. No. 2011/0009958, implant **500** in FIG. **5**B has spines **504** alternating with frames **506**. The spines and frames have curved cross-sections, and the circumferential extent of the spine cross-section is less than the circumferential extent of the frame cross-section. Implant **500** extends along a curved longitudinal axis, and its curvature plane bisects spines **504**. Each frame **506** has two struts **508** extending equally from the curvature plane and, therefore, equally from the spines on either side of that frame. Optional openings **510** are formed in each frame. Openings **510** communicate with a channel **532** extending along implant **500**. Channel **532** has an opening **534** along one side and extending through the spines and frames. An inlet portion **528** of implant **500** is configured to be disposed in the anterior chamber of the eye when rest of the implant is disposed in Schlemm's canal.

[0066] FIG. **6**A is a stylized line drawing illustrating histology slide HS shown in the previous figure. FIG. **6**B is a simplified cross-sectional view illustrating the eye from which the histology sample was taken. FIG. **6**A and FIG. **6**B are presented on a single page to illustrate the location of the histology sample relative to other portions of the eye **20**. As discussed earlier, eye **20** includes a dome-shaped wall **90** having a surface **92** defining an anterior chamber AC. Dome-shaped wall **90** of eye **20** comprises a cornea **36** and scleral tissue **34**. The scleral tissue **34** meets the cornea **36** at a limbus of eye **20**. In FIG. **6**B, surface **92** is shown having a generally hemispherical shape. In FIG. **6**A, ocular implant **500** is shown residing in Schlemm's canal SC.

[0067] In some embodiments, as shown in FIG. **2**, the implant is inserted from the anterior chamber through the trabecular meshwork into Schlemm's canal. Suitable delivery systems for this ab interno implantation procedure are described in US Publ. No. 2009/0132040, now U.S. Pat. Nos. 8,512,404; 8,337,509; US Publ. No. 2011/0098809; and U.S. application Ser. No. 13/330,592, filed

Dec. 19, 2011, now U.S. Pat. No. 8,663,150. As the implant passes through the relatively soft tissue of the trabecular meshwork, landing surfaces of the implant will engage the relatively stiffer scleral tissue bounding Schlemm's canal. The relative position of the landing surfaces with respect to other structure of the implant (and with respect to the structure of Schlemm's canal) will govern how reaction forces on the landing surfaces from this engagement will affect the implant as it is advanced into Schlemm's canal.

[0068] FIG. **7**A is a stylized line drawing showing a cross section of an ocular implant **500** residing in a section of an eye **20** including Schlemm's canal SC. FIG. **7**B is a section view showing same cross-sectional portion of ocular implant **500** prior to insertion into Schlemm' s canal of eye **20**. Implant **500** may be, e.g., one of the implants described in U.S. Pat. Nos. 7,740,604; 8,372,026; US 2009/0082860, now U.S. Pat. No. 8,734,377; or US 2011/0009958, now U.S. Pat. No. 8,425,449. In the view shown in FIG. **7**B, no external forces are acting on ocular implant **500** so that it is free to assume an undeformed curved shape free of stress and/or strain. FIG. **7**C is a perspective view showing the ocular implant **500** being inserted into Schlemm's canal through a cannula **72**. FIG. **7**A, FIG. **7**B and FIG. **7**C may be collectively referred to as FIG. **7**. Placing FIG. **7**A adjacent to FIG. **7**B allows comparisons to be drawn

[0069] therebetween. By comparing FIG. **7**A to FIG. **7**B, it will be appreciated that ocular implant **500** is assuming substantially the same orientation in both figures. It is believed that the tissues of the trabecular meshwork are quite soft and compliant, and they do not have sufficient stiffness to hold ocular implant **500** in a twisted orientation different from the orientation that the ocular implant takes when no external forces are acting on it. A twisted implant might therefore not remain in the desired position within Schlemm's canal.

[0070] Ocular implant **500** of FIG. **7**B includes a spine **504** and a frame **506**. In the exemplary embodiment of FIG. **7**B, frame **506** comprises a first strut **508**C and a second strut **508**D. As shown, struts **508**C and **508**D both extend circumferentially beyond the circumferential extent of spine **504**. First strut **508**C and second strut **508**D comprise a first landing surface **542**C and a second landing surface **542**D, respectively.

[0071] As shown in FIG. **7**C, the implant will engage Schlemm's canal tissue as it is inserted into and advanced along Schlemm's canal. The force vectors from engagement of the implant with scleral tissue partially bordering Schlemm's canal depend on the relative orientation of landing surfaces and scleral tissue with respect to the direction of insertion, which in tum depends on the orientation of the implant within the cannula and the relative positions of the landing surfaces on the implant. As shown in FIG. **7**C, during insertion of implant **500** into Schlemm's canal, landing surface **542**C is engaging the scleral tissue ST bordering Schlemm's canal more directly than its corresponding landing surface **542**D. This mismatch in engagement may result in the generation of different reaction forces on landing surfaces **542**C and **542**D and may apply bending or twisting moments to the implant that will twist implant **500** to move landing surface **542**D toward (and perhaps against) scleral tissue ST. The invention described below reduces this bending or twisting moment.

[0072] With reference to FIG. **7**, it will be appreciated that first landing surface **542**C of first strut **508**C and second landing surface **542**D of second strut **508**D define a footprint line **538**. Footprint line **538** contacts ocular implant **500** at a first point **540**C and a second point **540**D. First point **540**C is disposed on first landing surface **542**C of first strut **508**C. Second point **540**D is disposed on second landing surface **542**D of second strut **508**D.

[0073] A plane **524** is shown intersecting ocular implant **500** in FIG. **7**B. In the embodiment of FIG. **7**B, the longitudinal axis of ocular implant **500** follows a curved path so that the longitudinal axis defines a plane of curvature POC that bisects spines **504** and frames **506** and is co-planar with plane **524** shown in FIG. **7**B. A roll angle RA of frame **506** is illustrated using angular dimension lines in FIG. **7**B. Roll angle RA extends between plane **524** and footprint line **538**. In the embodiment of FIG. **7**B, because corresponding struts **508**C and **508**D extend equal distances from the plane of curvature, roll angle RA has a magnitude of about ninety degrees. Also in the embodiment of FIG. **7**B, footprint line **538** is generally orthogonal to the plane of curvature POC of ocular implant **500**.

[0074] As an ocular implant advances into Schlemm's canal during a delivery procedure, the ocular implant may press against the dome-shaped wall that defines the anterior chamber. More particularly, one or more struts of the ocular implant may press against scleral tissue forming part of the dome-shaped wall of the eye. Landing surfaces of the ocular implant may be seated against the outer wall of Schlemm's canal and the scleral tissue as it is advanced into Schlemm's canal. The scleral tissue may provide support for the outer wall of Schlemm's canal as the ocular implant is advanced therein. These supporting reaction forces will act against the ocular implant as the implant engages the wall of Schlemm's canal.

[0075] For example, as discussed above with respect to FIGS. **7**A and **7**C, as the ocular implant moves out of delivery cannula **72** into Schlemm's canal, first landing surface **542**C of first strut **508**C engages scleral tissue ST. Because of the slope of the inside back wall of Schlemm's canal, and the equal heights of the implant struts within the implant's plane of curvature, the impact force vector normal to the insertion path of the first landing surface **542**C against scleral tissue is greater than the impact force vector normal to the insertion path of the second landing surface **542**D against scleral tissue. The reaction forces from the scleral tissue against implant **500** may therefore cause a twisting or bending of implant **500** as it is advanced into Schlemm's canal that move landing surface **542**D toward, and perhaps against, scleral tissue ST.

[0076] Changing the angle of the insertion cannula with respect to the plane of Schlemm's canal would change the way the implant's landing surfaces interact with scleral tissue during insertion and, therefore, any bending or twisting moments applied to the implant. Visualization requirements and anterior chamber access limitations may require the cannula to form an angle greater than zero with the plane of Schlemm's canal. It therefore may be desirable to change the position of the implant landing surfaces in an effort to reduce the difference in the magnitude of the reaction forces on landing surfaces on opposite sides of the implant, such as by changing the relative heights of the struts with respect to the implant's plane of curvature.

[0077] FIG. **8**A is a stylized line drawing showing an ocular implant **100** residing in a section of an eye **20** including Schlemm's canal SC. FIG. **8**B is a section view showing a portion of ocular implant **100** prior to insertion into Schlemm's canal of eye **20**. FIG. **8**C is a perspective view showing the ocular implant **100** being inserted into Schlemm's canal through a cannula **72**. In the view shown in FIG. **8**B, no external forces are acting on ocular implant **100** so that it is free to assume an undeformed shape free of stress and/or strain. By comparing FIG. **8**A to FIG. **8**B, it will be appreciated that ocular implant **100** is assuming substantially the same orientation in both figures. As mentioned above, the tissues of the trabecular meshwork are quite soft and compliant, and they and do not have sufficient stiffness to hold ocular implant **100** in a twisted orientation different from the orientation that the ocular implant takes when no external forces are acting on it.

[0078] Ocular implant **100** of FIG. **8**B includes a spine **104** and a frame **106**. In the exemplary embodiment of FIG. **8**B, frame **106** comprises a first strut **108**C and a second strut **108**D. As shown, struts **108**C and **108**D both extend circumferentially beyond the circumferential extent of spine **104**. First strut **108**C and second strut **108**D comprise a first landing surface **142**C and a second landing surface **142**D, respectively. With reference to FIG. **8**, it will be appreciated that first landing surface **142**C of first strut **108**C and second landing surface **142**D of second strut **108**D define a footprint line **138**. Footprint line **138** contacts ocular implant **100** at a first point **140**C and a second point **140**D. First point **140**C is disposed on first landing surface **142**C of first strut **108**C. Second point **140**D is disposed on second landing surface **142**D of second strut **108**D.

[0079] A plane **124** is shown intersecting ocular implant **100** in FIG. **8**B. In the embodiment of FIG. **8**B, the longitudinal axis of ocular implant **100** follows a curved path so that the longitudinal axis defines a plane of curvature POC that is co-planar with plane **124** shown in FIG. **8**B. As shown in FIG. **8**B, the plane of curvature POC bisects spine **104** so that the cross-sectional circumferential extent of spine **104** on one side of the plane of curvature is substantially equal to the cross-sectional circumferential extent of spine **104** on the other side of the plane of curvature.

[0080] During implantation, the implant **100** will be oriented with the curved insertion cannula **72**, as suggested by FIG. **8**C, such that the implant's plane of curvature is coplanar with the cannula's plane of curvature. Thus, adjusting the height of the struts with respect to the implant's plane of curvature and curved longitudinal axis will affect the way that the implant's landing surfaces engage scleral tissue as the implant is advanced into and along Schlemm's canal.

[0081] A roll angle RB of frame **106** is illustrated using angular dimension lines in FIG. **8**B. Roll angle RB extends between plane **124** and footprint line **138**. By comparing FIG. **8**B with FIG. **7**B described above, it will be appreciated that because strut **108**D extends further out of the plane of curvature than its opposing strut **108**C, the roll angle RB shown in FIG. **8**B has a magnitude different from the magnitude of roll angle RA shown in FIG. **7**B. In the embodiment of FIG. **8**B, angle RB has a magnitude other than ninety degrees. Also in the embodiment of FIG. **8**B, footprint line **138** is generally skewed relative to the plane of curvature POC of ocular implant **100**.

[0082] Differences in the responses of implant **100** and implant **500** can be seen by comparing FIG. **8** to FIG. **7**. As shown in FIGS. **8**A and **8**C, as the ocular implant **100** moves out of delivery cannula **72** into Schlemm's canal, landing surfaces **142**C and **142**D engage scleral tissue ST. Because of the slope of the inside back wall of Schlemm's canal and the unequal heights of the implant struts with respect to the implant's plane of curvature, the impact force vector normal to the insertion path of the first landing surface **142**C against scleral tissue is closer to (and possibly equal to) the impact force vector normal to the insertion path of the second landing surface **142**D against scleral tissue. The decrease in the difference between the reaction forces on opposite sides of the implant will decrease any bending or twisting moments applied to implant **100** during insertion and advancement within Schlemm's canal.

[0083] FIG. **9** is a perspective view showing ocular implant **500** of FIG. **7**. With reference to FIG. **9**, it will be appreciated that ocular implant **500** defines a cylindrical surface CYL enclosing a three dimensional volume having a shape similar to a cylinder. Ocular implant **500** of FIG. **9** includes a spine **504** and a frame **506**. In the exemplary embodiment of FIG. **9**, frame **506** comprises a first strut **508**C and a second strut **508**D. First strut **508**C and second strut **508**D comprise a first landing surface **542**C and a second landing surface **542**D, respectively. With reference to FIG. **7**, it will be appreciated that first landing surface **542**C of first strut **508**C and second landing surface **542**D of second strut **508**D define a footprint line **538**. Footprint line **538** contacts ocular implant **500** at a first point **540**C and a second point **540**D. First point **540**C is disposed on first landing surface **542**C of first strut **508**C. Second point **540**D is disposed on second landing surface **542**D of second strut **508**D. With reference to FIG. **9**, it will be appreciated that footprint line **538** lies on the cylindrical surface CYL defined by ocular implant **500**.

[0084] FIG. **10** is an additional perspective view showing ocular implant **100** of FIG. **8**. With reference to FIG. **10**, it will be appreciated that ocular implant **100** defines a conical surface C enclosing a three dimensional volume having a shape similar to a cone. Ocular implant **100** of FIG. **10** includes a spine **104** and a frame **106**. In the exemplary embodiment of FIG. **10**, frame **106** comprises a first strut **108**C and a second strut **108**D. First strut **108**C and second strut **108**D comprise a first landing surface **142**C and a second landing surface **142**D, respectively. With reference to FIG. **7**, it will be appreciated that first landing surface **142**C of first strut **108**C and second landing surface **142**D of second strut **108**D define a footprint line **138**. Footprint line **138** contacts ocular implant **100** at a first point **140**C and a second point **140**D. First point **140**C is disposed on first landing surface **142**C of first strut **108**C. Second point **140**D is disposed on second landing surface **142**D of second strut **108**D. With reference to FIG. **10**, it will be appreciated that footprint line **138** lies on the conical surface C defined by ocular implant **100**.

[0085] FIG. **11**A is a stylized perspective view showing a conical surface C that is sized and

positioned so as to intersect a hemispherical surface Sin two places. A first line LI is formed where the two surfaces intersect a first time. A second line L**2** is formed where conical surface C and hemispherical surface S intersect a second time. A first point **142**A is positioned on first line LI and a second point **142**B is disposed on second line L**2**.
[0086] FIG. **11**B is a stylized perspective view showing the ocular implant **100** of FIG. **8** disposed inside a chamber V defined by a hemispherical surface S. Ocular implant **100** contacts hemispherical surface S at a first point **140**A, a second point **140**B, a third point **140**C, a fourth point **140**D, a fifth point **140**E, and a sixth point **140**F. First point **140**A is disposed on a first landing surface **142**A of ocular implant **100**. Second point **140**B is disposed on a second landing surface **142**B of ocular implant **100**. Third point **140**C and fourth point **140**D are disposed on a third landing surface **142**C and a fourth landing surface **142**D, respectively. Fifth point **140**E and sixth point **140**F are disposed on a fifth landing surface **142**E and a sixth landing surface **142**F, respectively.
[0087] Applicant has created ocular implants designed to work in harmony with the dome shaped wall that defines the anterior chamber of the human eye. In some useful embodiments, the ocular implants are configured such that reaction forces applied to the ocular implant by scleral tissue while the ocular implant is being advanced into Schlemm's canal subject the ocular implant to pure bending with little or no twisting. The ocular implant may be configured such that a net twisting moment applied to each spine by the normal forces supporting the landing surfaces is substantially zero. The ocular implant may also be configured such that the normal forces supporting the landing surfaces primarily or exclusively act to bend each spine along the preferential bending plane thereof.
[0088] The implant will bend preferentially about the region having the smallest circumferential extent, i.e., the spine. In the embodiment shown FIGS. **8**, **10** and **11**B, because the spines extend equally on both sides of the plane of curvature POC, the plane of curvature and the preferential bending plane are coplanar. If the roll angle of the implant is set so that the normal forces on the strut landing surfaces are substantially equal (i.e., the angle of the footprint equals the angle of the back wall of Schlemm's canal), the net forces on the implant during insertion and advancement will cause the implant to bend along its preferential bending plane with no net twisting about the plane. In some other useful embodiments (discussed below with respect to FIGS. **15-18** and **19**B), the preferential bending plane of each spine extends in a direction that is at right angles to a conical surface defined by the ocular implant.
[0089] In the embodiment of FIG. **11**B, ocular implant **100** defines a conical surface C. Ocular implant **100** contacts conical surface C at a first point **140**A, a second point **140**B, a third point **140**C, a fourth point **140**D, a fifth point **140**E, and a sixth point **140**F. Due to page size constraints, conical surface C is truncated in FIG. **21**. Conical surface C intersects hemispherical surface S in two places. First point **140**A, third point **140**C, and fifth point **140**E are disposed on a first line formed where conical surface C and hemispherical surface S intersect a first time. Second point **140**B, fourth point **140**D, and sixth point **140**F are disposed on a second line formed where the two surfaces intersect a second time.
[0090] FIG. **12**A, FIG. **12**B and FIG. **12**C are plan views of the ocular implant **100** of FIG. **8** created using multiview projection. FIG. **12**A, FIG. **12**B and FIG. **12**C may be referred to collectively as FIG. **12**. It is customary to refer to multiview projections using terms such as front view, top view, and side view. In accordance with this convention, FIG. **12**A may be referred to as a top view of ocular implant **100**, FIG. **12**B may be referred to as a side view of ocular implant **100**, and FIG. **12**C may be referred to as a bottom view of ocular implant **100**. The terms top view, side view, and bottom view are used herein as a convenient method for differentiating between the views shown in FIG. **12**. It will be appreciated that the implant shown in FIG. **12** may assume various orientations without deviating from the spirit and scope of this detailed description. Accordingly, the terms top view, side view, and bottom view should not be interpreted to limit the

scope of the invention recited in the attached claims.

[0091] Ocular implant **100** of FIG. **12** comprises a body **102** that extends along a longitudinal central axis **120**. In the exemplary embodiment of FIG. **12**, longitudinal central axis **120** follows a curved path such that longitudinal central axis **120** defines a curvature plane **122**. In some useful embodiments, the radius of curvature of the ocular implant is substantially equal to the radius of curvature of Schlemm's canal. Body **102** of ocular implant **100** has a distal end **126**, a proximal inlet portion **128** and an intermediate portion **130** extending between the proximal inlet portion **128** and the distal end **126**. Intermediate portion **130** comprises a plurality of spines **104** and a plurality of frames **106**. The spines **104** of intermediate portion **130** include a first spine **104**A, a second spine **104**B and a third spine **104**C. The frames **106** of intermediate portion **130** include a first frame **106**A, a second frame **106**B and a third frame **106**C. Ocular implant **100** is sized and configured so that the spines and frames are disposed in and supporting Schlemm's canal and the inlet **128** is disposed in the anterior chamber to provide for flow of aqueous humor from the anterior chamber through Schlemm's canal to outflow channels communicating with Schlemm's canal.

[0092] In FIG. **12**, first spine **104**A can be seen extending distally beyond proximal inlet portion **128**. First frame **106**A comprises a first strut **108**A and a second strut **108**B that extend between first spine **104**A and second spine **104**B. With reference to FIG. **12**, it will be appreciated that second frame **106**B abuts a distal end of second spine **104**B. In the embodiment of FIG. **12**, second frame **106**B comprises a first strut **108**C and a second strut **108**D that extend between second spine **104**B and third spine **104**C. Third frame **106**C can be seen extending between third spine **104**C and distal end **126** of ocular implant **100** in FIG. **12**. Third frame **106**C comprises a first strut **108**E and a second strut **108**F.

[0093] Body **102** of ocular implant **100** defines a channel **132** that opens into a channel opening **134**. With reference to FIG. **12**, it will be appreciated that channel **132** and channel opening **134** extending together through body **102** across first spine **104**A, second spine **104**B, third spine **104**C, first frame **106**A, second frame **106**B, and third frame **106**C. Optional additional openings **110** communicating with channel **132** are disposed between the spines and are surrounded by the struts.

[0094] With particular reference to FIG. **12**B, it will be appreciated that curvature plane **122** intersects first spine **104**A, second spine **104**B, and third spine **104**C. In the embodiment of FIG. **12**A, curvature plane **122** bisects each spine into two halves. The two halves of each spine are symmetrically shaped about curvature plane **122** in the embodiment of FIG. **12**A. With reference to FIG. **12**B and FIG. **12**C, it will be appreciated that the frames of ocular implant **100** are not symmetric about curvature plane **122**.

[0095] FIG. **13**A is a plan view showing ocular implant **100** of FIG. **8**. FIG. **13**B is an enlarged section view taken along section line B-B shown in FIG. **13**A. FIG. **13**C is an additional enlarged section view taken along section line C-C shown in FIG. **13**A. FIG. **13**A, FIG. **13**B and FIG. **13**C may be collectively referred to as FIG. **13**.

[0096] Ocular implant **100** of FIG. **13** comprises a body **102** that extends along a longitudinal central axis **120**. Body **102** of ocular implant **100** has a distal end **126**, a proximal inlet portion **128** and an intermediate portion **130** extending between the proximal inlet portion **128** and the distal end **126**. Intermediate portion **130** comprises a plurality of spines **104** and a plurality of frames **106**. The spines **104** of intermediate portion **130** include a first spine **104**A, a second spine **104**B and a third spine **104**C. The frames **106** of intermediate portion **130** include a first frame **106**A, a second frame **106**B and a third frame **106**C.

[0097] In FIG. **13**, first spine **104**A can be seen extending distally beyond proximal inlet portion **128**. First frame **106**A comprises a first strut **108**A and a second strut **108**B that extend between first spine **104**A and second spine **104**B. With reference to FIG. **13**, it will be appreciated that second frame **106**B abuts a distal end of second spine **104**B. In the embodiment of FIG. **13**, second frame **106**B comprises a first strut **108**C and a second strut **108**D that extend between second spine

104B and third spine **104**C. Third frame **106**C can be seen extending between third spine **104**C and distal end **126** of ocular implant **100** in FIG. **13**. Third frame **106**C comprises a first strut **108**E and a second strut **108**F.

[0098] Body **102** of ocular implant **100** defines a channel **132** that opens into a channel opening **134**. With particular reference to FIG. **13**A, it will be appreciated that channel **132** and channel opening **134** extending together through body **102** across first spine **104**A, second spine **104**B, third spine **104**C, first frame **106**A, second frame **106**B, and third frame **106**C. In this embodiment, the struts on one side of the implant extend further out of the plane of curvature that their corresponding struts on the opposite side of the implant. Thus, as shown in FIG. **13**B, strut **108**F extends further out of the plane of curvature (corresponding with longitudinal central axis **120**) than its opposing strut **108**E.

[0099] FIG. **14**A is a perspective view showing ocular implant **100** of FIG. **8**. Ocular implant **100** comprises a body **102** extending along a longitudinal central axis **120**. A first plane **124**A is shown intersecting ocular implant **100** in FIG. **14**A. In the embodiment of FIG. **14**A, longitudinal central axis **120** follows a path that is generally curved such that longitudinal central axis **120** defines a plane of curvature that is co-planar with first plane **124**A shown in FIG. **14**A. A second plane **124**B, a third plane **124**C, and a fourth plane **124**D are also shown intersecting ocular implant **100** in FIG. **14**A. Second plane **124**B, third plane **124**C, and fourth plane **124**D are all transverse to ocular implant **100** and longitudinal central axis **120** in the embodiment of FIG. **14**A. More particularly, in the exemplary embodiment of FIG. **14**A, second plane **124**B is orthogonal to a reference line **136**B that lies in first plane **124**A and is tangent to longitudinal central axis **120**. Third plane **124**C is orthogonal to a reference line **136**C that lies in first plane **124**A and is tangent to longitudinal central axis **120**. Third plane **124**D is orthogonal to a reference line **136**D that lies in first plane **124**A and is tangent to longitudinal central axis **120**.

[0100] Body **102** of ocular implant **100** has a distal end **126**, a proximal inlet portion **128** and an intermediate portion **130** extending between the proximal inlet portion **128** and the distal end **126**. Intermediate portion **130** comprises a plurality of spines **104** and a plurality of frames **106**. The frames **106** of intermediate portion **130** include a first frame **106**A, a second frame **106**B and a third frame **106**C. In FIG. **14**A, second plane **124**B is shown extending through first frame **106**A. Third plane **124**C and fourth plane **124**D are shown extending through second frame **106**B and third frame **106**C, respectively, in FIG. **14**A.

[0101] The spines **104** of intermediate portion **130** include a first spine **104**A, a second spine **104**B and a third spine **104**C. In FIG. **14**, first spine **104**A can be seen extending distally beyond proximal inlet portion **128**. First frame **106**A comprises a first strut **108**A and a second strut **108**B that extend between first spine **104**A and second spine **104**B. With reference to FIG. **14**, it will be appreciated that second frame **106**B abuts a distal end of second spine **104**B. In the embodiment of FIG. **14**, second frame **106**B comprises a third strut and a fourth strut that extend between second spine **104**B and third spine **104**C. Third frame **106**C can be seen extending between third spine **104**C and distal end **126** of ocular implant **100** in FIG. **14**. Third frame **106**C comprises a fifth strut and a sixth strut.

[0102] With reference to FIG. **14**A, it will be appreciated that first plane **124**A intersects the spines of ocular implant **100**. In the embodiment of FIG. **14**A, first plane **124**A bisects each spine into two halves. The two halves of each spine are symmetrically shaped about first plane **124**A in the embodiment of FIG. **14**A.

[0103] In the exemplary embodiment of FIG. **14**A, each frame **106** comprises two struts. In some useful embodiments, each strut includes a landing surface and each frame is configured to support the spines in a location offset from an outer major side of Schlemm's canal while the ocular implant is in Schlemm's canal and the landing surfaces are engaging the outer major side of Schlemm's canal. In the embodiment of FIG. **14**A, first frame **106**A, second frame **106**B and third frame **106**C are each oriented at a roll angle.

[0104] First frame **106**A of FIG. **14**A comprises a first strut **108**A and a second strut **108**B. The roll angle of first frame **106**A may be defined by a footprint line defined by a first landing surface of first strut **108**A and a second landing surface of second strut **108**B. In the embodiment of FIG. **14**A, the footprint line will lie in second plane **124**B and be skewed relative to first plane **124**A. The angle between the footprint line and first plane **124**A may be referred to as the roll angle of the frame. In the exemplary embodiment of FIG. **14**A, second frame **106**B and third frame **106**C have roll angles similar to the roll angle of first frame **106**A.

[0105] FIG. **14**B is a plan view further illustrating first frame **106**A and second plane **124**B shown in FIG. **14**A. With reference to FIG. **14**B, it will be appreciated that first frame **106**A has a lateral cross-sectional shape F that lies in second plane **124**B. First frame **106**A comprises a first strut **108**A and a second strut **108**B. First plane **124**A is shown intersecting first frame **106**A in FIG. **14**B.

[0106] A roll angle RA of first frame **106**A is illustrated using angular dimension lines in FIG. **14**B. Roll angle RA extends between first plane **124**A and a first footprint line **138**A. In the exemplary embodiment of FIG. **14**, first footprint line **138**A is defined by a first point **140**A and a second point **140**B. First point **140**A is disposed on a first landing surface **142**A of first strut **108**A. Second point **140**B is disposed on a second landing surface **142**B of second strut **108**B. As discussed above, in this embodiment the struts on one side of the implant extend further out of the plane of curvature than their corresponding struts on the opposite side of the implant. Thus, as shown in FIG. **14**B, strut **108**B extends further out of the plane of curvature **124**A than its opposing strut **108**A.

[0107] FIG. **15**A, FIG. **15**B and FIG. **15**C are multi-plan views of yet another exemplary ocular implant **300** in accordance with the present detailed description. FIG. **15**A, FIG. **15**B and FIG. **15**C may be referred to collectively as FIG. **15**. It is customary to refer to multi-view projections using terms such as front view, top view, and side view. In accordance with this convention, FIG. **15**A may be referred to as a top view of ocular implant **300**, FIG. **15**B may be referred to as a side view of ocular implant **300**, and FIG. **15**C may be referred to as a bottom view of ocular implant **300**. The terms top view, side view, and bottom view are used herein as a convenient method for differentiating between the views shown in FIG. **15**. It will be appreciated that the implant shown in FIG. **15** may assume various orientations without deviating from the spirit and scope of this detailed description. Accordingly, the terms top view, side view, and bottom view should not be interpreted to limit the scope of the invention recited in the attached claims.

[0108] Ocular implant **300** of FIG. **15** comprises a body **302** that extends along a longitudinal central axis **320**. In the exemplary embodiment of FIG. **15**, longitudinal central axis **320** follows a curved path such that longitudinal central axis **320** defines a curvature plane **322**. Body **302** of ocular implant **300** has a distal end **326**, a proximal inlet portion **328** and an intermediate portion **330** extending between the proximal inlet portion **328** and the distal end **326**. Intermediate portion **330** comprises a plurality of spines **304** and a plurality of frames **306**. The spines **304** of intermediate portion **330** include a proximal-most spine **304**A, an intermediate spine **304**B and a distal-most spine **304**C. The frames **306** of intermediate portion **330** include a proximal-most frame **306**A, an intermediate frame **306**B and a distal-most frame **306**C. Ocular implant **300** is sized and configured so that the spines and frames are disposed in and supporting Schlemm's canal and the inlet **328** is disposed in the anterior chamber to provide for flow of aqueous humor from the anterior chamber through Schlemm's canal to outflow channels communicating with Schlemm's canal.

[0109] In FIG. **15**, proximal-most spine **304**A can be seen extending distally beyond proximal inlet portion **328**. Proximal-most frame **306**A comprises a first strut **308**A and a second strut **308**B that extend between proximal-most spine **304**A and intermediate spine **304**B. With reference to FIG. **15**, it will be appreciated that intermediate frame **306**B abuts a distal end of intermediate spine **304**B. In the embodiment of FIG. **15**, intermediate frame **306**B comprises a third strut **308**C and a fourth strut **308**D that extend between intermediate spine **304**B and distal-most spine **304**C. Distal-most frame **306**C can be seen extending between distal-most spine **304**C and distal end **326** of ocular implant **300** in FIG. **15**. Distal-most frame **306**C comprises a fifth strut **308**E and a sixth strut **308**F.

[0110] Body **302** of ocular implant **300** defines a channel **332** that opens into a channel opening **334**. With reference to FIG. **15**, it will be appreciated that channel **332** and channel opening **334** extending together through body **302** across proximal-most spine **304**A, intermediate spine **304**B, distal-most spine **304**C, proximal-most frame **306**A, intermediate frame **306**B, and distal-most frame **306**C. Optional additional openings **310** communicating with channel **332** are disposed between the spines and are surrounded by the struts.

[0111] FIG. **16**A is a plan view showing ocular implant **300** of FIG. **15**. FIG. **16**B is an enlarged section view taken along section line B-B shown in FIG. **16**A. FIG. **16**C is an additional enlarged section view taken along section line C-C shown in FIG. **16**A. FIG. **16**A, FIG. **16**B and FIG. **16**C may be collectively referred to as FIG. **16**.

[0112] Ocular implant **300** of FIG. **16** comprises a body **302** that extends along a longitudinal central axis **320** which, in this view, lies in the plane of curvature of implant **300**. Body **302** of ocular implant **300** has a distal end **326**, a proximal inlet portion **328** and an intermediate portion **330** extending between the proximal inlet portion **328** and the distal end **326**. Intermediate portion **330** comprises a plurality of spines **304** and a plurality of frames **306**. The spines **304** of intermediate portion **330** include a proximal-most spine **304**A, an intermediate spine **304**B and a distal-most spine **304**C. The frames **306** of intermediate portion **330** include a proximal-most frame **306**A, an intermediate frame **306**B and a distal-most frame **306**C. As shown, unlike the embodiment of FIG. **8**, in this embodiment the plane of curvature does not bisect the spines.

[0113] In FIG. **16**, proximal-most spine **304**A can be seen extending distally beyond proximal inlet portion **328**. Proximal-most frame **306**A comprises a first strut **308**A and a second strut **308**B that extend between proximal-most spine **304**A and intermediate spine **304**B. With reference to FIG. **16**, it will be appreciated that intermediate frame **306**B abuts a distal end of intermediate spine **304**B. In the embodiment of FIG. **16**, intermediate frame **306**B comprises a third strut **308**C and a fourth strut **308**D that extend between intermediate spine **304**B and distal-most spine **304**C. Distal-most frame **306**C can be seen extending between distal-most spine **304**C and distal end **326** of ocular implant **300** in FIG. **16**. Distal-most frame **306**C comprises a fifth strut **308**E and a sixth strut **308**F.

[0114] Body **302** of ocular implant **300** defines a channel **332** that opens into a channel opening **334**. With reference to FIG. **16**, it will be appreciated that channel **332** and channel opening **334** extending together through body **302** across proximal-most spine **304**A, intermediate spine **304**B, distal-most spine **304**C, proximal-most frame **306**A, intermediate frame **306**B, and distal-most frame **306**C. In this embodiment, the struts on one side of the implant extend further out of the plane of curvature (shown as a dotted line in FIG. **16**B) than their corresponding struts on the opposite side of the implant. Thus, as shown in FIG. **16**B, strut **308**F extends further out of the plane of curvature than its opposing strut **308**E. It can also be seen from FIG. **16**B that spine **304**C extends further out of the plane of curvature on one side than on the other and that the struts **308**E and **308**F both extend circumferentially equally beyond the circumferential extend of spine **304**C. Thus, implant **300** does not bend preferentially in the implant's plane of curvature.

[0115] FIG. **17**A is a perspective view showing ocular implant **300** of FIGS. **15** and **16**. A first plane **324**A is shown intersecting ocular implant **300** in FIG. **17**A. Ocular implant **300** of FIG. **17**A comprises a body **302** extending along a longitudinal central axis **320**. In the embodiment of FIG. **17**A, longitudinal central axis **320** follows a path that is generally curved such that longitudinal central axis **320** defines a plane of curvature POC.

[0116] In the embodiment of FIG. **17**A, plane of curvature POC is co-planar with first plane **324**A shown in FIG. **17**A. With reference to FIG. **17**A, it will be appreciated that plane of curvature POC would no longer be coplanar with first plane **324**A if ocular implant **300** was rotated. In some methods in accordance with this detailed description, ocular implant may be advanced into Schlemm's canal while the plane of curvature of the ocular implant is co-planar with a plane of curvature of Schlemm's canal.

[0117] A second plane **324**B and a third plane **324**C are shown extending transversely across body **302** of ocular implant **300** in FIG. **17**A. With reference to FIG. **17**A, it will be appreciated that third plane **324**C extends through a distal-most spine **304**C of ocular implant **300**. Second plane **324**B is shown extending through a distal-most frame **306**C of ocular implant **300** in FIG. **17**A. In the exemplary embodiment of FIG. **17**A, second plane **324**B is orthogonal to a reference line **336**B. Reference line **336**B is tangent to longitudinal central axis **320** and is shown lying on first plane **324**A in FIG. **17**A.

[0118] Body **302** of ocular implant **300** has a distal end **326**, a proximal inlet portion **328** and an intermediate portion **330** extending between the proximal inlet portion **328** and the distal end **326**. Intermediate portion **330** comprises a plurality of spines **304** and a plurality of frames **306**. The frames **306** of intermediate portion **330** include a proximal-most frame **306**A, an intermediate frame **306**B and a distal-most frame **306**C. Second plane **324**B is shown extending through distal-most frame **306**C in FIG. **17**A. In the exemplary embodiment of FIG. **17**A, body **302** includes a single intermediate frame **306**B. It will be appreciated, however, that body **302** include any number of intermediate frames without deviating from the spirit and scope of the present detailed description.

[0119] The spines **304** of intermediate portion **330** include a proximal-most spine **304**A, an intermediate spine **304**B and a distal-most spine **304**C. In FIG. **17**A, third plane **324**C is shown extending through distal-most spine **304**C. In the exemplary embodiment of FIG. **17**A, body **302** includes a single intermediate spine **304**B. It will be appreciated, however, that body **302** include any number of intermediate spines without deviating from the spirit and scope of the present detailed description.

[0120] In FIG. **17**A, proximal-most spine **304**A can be seen extending distally beyond proximal inlet portion **328**. Proximal-most frame **306**A comprises a first strut **308**A and a second strut **308**B that extend between proximal-most spine **304**A and intermediate spine **304**B. With reference to FIG. **17**, it will be appreciated that intermediate frame **306**B abuts a distal end of intermediate spine **304**B. In the embodiment of FIG. **17**, intermediate frame **306**B comprises a first strut and a second strut that extend between intermediate spine **304**B and distal-most spine **304**C. Distal-most frame **306**C can be seen extending between distal-most spine **304**C and distal end **326** of ocular implant **300** in FIG. **17**. Distal-most frame **306**C comprises a first strut **308**E and a second strut **308**F. Second plane **324**B is shown extending through distal-most frame **306**C in FIG. **17**A.

[0121] FIG. **17**B is an enlarged plan view of second plane **324**B shown in FIG. **17**A. With reference to FIG. **17**A, it will be appreciated that second plane **324**B intersects a first strut **308**E and a second strut **308**F of distal-most frame **306**C. In FIG. **17**B, a profile PE of first strut **308**E is shown lying on second plane **324**B. A profile PF of second strut **308**F is also shown lying on second plane **324**B in FIG. **17**B. The profile of each strut is filled with a cross-hatch pattern in FIG. **17**B.

[0122] In the embodiment of FIG. **17**, the body of ocular implant **300** extends along a longitudinal central axis that is generally curved such that the longitudinal central axis defines a plane of curvature POC that is represented by a dashed line in FIG. **17**B. A roll angle RA of distal-most frame **306**C is illustrated using angular dimension lines in FIG. **17**B. Roll angle RA extends between plane of curvature POC and a first footprint line **338**C. In the exemplary embodiment of FIG. **17**, first footprint line **338**C is defined by a first point **340**E and a second point **340**F. First point **340**E is disposed on a first landing surface **342**E of first strut **308**E. Second point **340**F is disposed on a second landing surface **342**F of second strut **308**F.

[0123] Upon advancement of ocular implant **300** into Schlemm's canal, and depending on the angle of the delivery cannula with respect to the plane of Schlemm's canal during insertion, first landing surface **342**E of first strut **308**E and second landing surface **342**F of second strut **308**F may seat against the inner surface of the dome shaped wall that encloses the anterior chamber with the dome shaped wall providing normal forces supporting the landing surfaces. In some useful embodiments, roll angle RA is selected such that, when ocular implant **300** is advanced into Schlemm's canal landing surfaces of first strut **308**E and second strut **308**F are seated against the dome-shaped wall that defines the anterior chamber of the eye with substantially equal force. The decrease in the difference between the reaction forces on opposite sides of the implant will decrease any bending or twisting moments applied to implant **300** during insertion and advancement within Schlemm's canal.

[0124] FIG. **17**C is an enlarged plan view of third plane **324**C shown in FIG. **17**A. As shown in FIG. **17**A, third plane **324**C intersects distal-most spine **304**C. Accordingly, distal-most spine **304**C is shown in cross-section in FIG. **17**C. Distal-most spine **304**C has a profile LC that is shown lying on third plane **324**C in FIG. **17**C.

[0125] In the embodiment of FIG. **17**, the body of ocular implant **300** extends along a longitudinal central axis that is generally curved such that the longitudinal central axis defines a plane of curvature POC that is represented by a dashed line in FIG. **17**C. Because the plane of curvature POC does not bisect spine **304**C, spine **304**C will not bend preferentially about POC.

[0126] Upon advancement of ocular implant **300** into Schlemm's canal, the landing surfaces may seat against the inner surface of the dome shaped wall that encloses the anterior chamber with the dome shaped wall providing normal forces supporting the landing surfaces. Each spine of ocular implant **300** may be configured to preferentially bend along a preferential bending plane. In some useful embodiments, each spine is rotationally offset relative to a first adjacent frame and a second adjacent frame by an angle selected such that the normal forces supporting the landing surfaces primarily or exclusively act to bend each spine along the preferential bending plane thereof. In some useful embodiments, each spine is rotationally offset relative to a first adjacent frame and a second adjacent frame by an angle selected such that a net twisting moment applied to each spine by the normal forces is substantially zero. The arrangement described above may minimize any twisting of the ocular implant body as the ocular implant is advanced into Schlemm's canal as part of a delivery procedure. This arrangement may also provide better trackability than devices that do not include these design features.

[0127] As shown in FIG. **17**C, distal-most spine **304**C of ocular implant **300** has a first lateral extent EF and a second lateral extent ES. In some useful embodiments, an aspect ratio of first lateral extent EF to second lateral extent ES is greater than about one. In some useful embodiments, the aspect ratio of first lateral extent EF to second lateral extent ES is greater than about three. The relationships described above may advantageously cause distal-most spine **304**C to preferential bend more along one direction over another by, e.g., bending about the thinnest portion of the device.

[0128] With reference to FIG. **17**C, it will be appreciated that distal-most spine **304**C has a thickness T. In some useful embodiments, an aspect ratio of first lateral extent EF to thickness T may be selected such that distal-most spine **304**C preferentially bends more along one direction over another. In some useful embodiments, an aspect ratio of first lateral extent EF to thickness T is greater than about one. In some useful embodiments, the aspect ratio of first lateral extent EF to thickness T is greater than about three.

[0129] FIG. **18**A is an additional perspective view of ocular implant **300** shown in the previous figure. Ocular implant **300** of FIG. **18**A includes a distal-most spine **304**C and a distal-most frame **306**C. In the exemplary embodiment of FIG. **18**A, distal-most frame **306**C comprises a first strut **308**E and a second strut **308**F. FIG. **18**B is a stylized isometric view showing the profiles of distal-most spine **304**C, first strut **308**E and second strut **308**F. The profile of distal-most spine **304**C was created where a third plane **324**C intersects ocular implant **300**. Similarly, the profiles of first strut **308**E and second strut **308**F were created where a second plane **324**B intersects ocular implant **300** in FIG. **18**A.

[0130] The profiles of first strut **308**E and second strut **308**F are labeled PE and PF in FIG. **18**B. The profile of distal-most spine **304**C is labeled LC in FIG. **18**B. With reference to those profiles, it will be appreciated that first strut **308**E and second strut **308**F comprise a first landing surface **342**E and a second landing surface **342**F, respectively. In FIG. **18**B, a first normal force FE is represented by an arrow that is shown contacting first landing surface **342**A. A second normal force FF is represented by an arrow that is shown contacting second landing surface **342**A in FIG. **18**.

[0131] Distal-most spine **304**C of FIG. **18**, is configured to preferentially bend in a preferential bending direction D shown in FIG. **18**B. In some useful embodiments, ocular implant **300** is configured so that direction D extends at right angles to a point on a spherical surface that defines the anterior chamber when the landing surfaces of the ocular implant are seated against the dome-shaped wall that defines the anterior chamber. Also in some useful embodiments, ocular implant **300** is configured so that direction D extends at right angles to a point on a conical surface defined by ocular implant **300**. With reference to FIG. **18**, it will be appreciated that direction D is generally parallel to the directions of first normal force vector FE and second normal force vector FF.

[0132] In the embodiment of FIG. **18**, first strut **308**E, second strut **308**F and distal-most spine **304**C are configured such that, when ocular implant **300** is advanced along Schlemm's canal as part of a delivery procedure the landing surfaces of first strut **308**E and second strut **308**F will be seated against the dome-shaped wall defining the anterior chamber. When first landing surface **342**A and second landing surface **342**B contact the outer major wall of Schlemm's canal, the dome-shaped wall provides normal forces to support first strut **308**E and second strut **308**F. In the embodiment of FIG. **18**, ocular implant **300** is configured such that normal forces applied to the landing surfaces of first strut **308**E and second strut **308**F primarily or exclusively act to bend first spine **304**C along its preferential bending plane PBP. In some useful embodiments, each spine is rotationally offset relative to a first adjacent frame and a second adjacent frame by an angle selected such that the normal forces supporting the landing surfaces of the frames primarily or exclusively act to bend each spine along the preferential bending plane thereof. In some useful embodiments, each spine is rotationally offset relative to a first adjacent frame and a second adjacent frame by an angle selected such that a net twisting moment applied to each spine by the normal forces is substantially zero.

[0133] FIG. **19**A and FIG. **19**B are perspective views showing distal portions ocular implant **100** of FIG. **8** and ocular implant **300** of FIG. **15**, respectively. FIG. **19**A and FIG. **19**B are presented on a single page so that second ocular implant **300** can be easily compared and contrasted to first ocular implant **100**. FIG. **19**A and FIG. **19**B may be collectively referred to as FIG. **19**. In FIG. **19**, the body of each ocular implant extends along a longitudinal central axis that is generally curved such that the longitudinal central axis defines a plane of curvature POC. Each plane of curvature POC is represented by a dashed line in FIG. **19**.

[0134] Ocular implant **100** includes a distal-most spine **104**C, and ocular implant **300** includes a distal-most spine **304**C. As shown in FIG. **19**A, plane of curvature POC bisects spine **104**C. Spine **104**C will bend preferentially about POC. In the embodiment of FIG. **19**B, the plane of curvature POC does not bisect distal-most spine **304**C of ocular implant **300**. Spine **304**C will not bend preferentially in plane POC.

[0135] In FIG. **19**, two normal forces are shown acting on each ocular implant. A first normal force FE is represented by an arrow that is shown contacting a first landing surface of each ocular implant. A second normal force FF is represented by an arrow that is shown contacting second landing surface of each ocular implant. The arrows representing first normal force FE and second normal force FF are force vectors representing reaction forces provided by the dome-shaped wall of the eye. The dome-shaped wall of the eye provides support for the outer major wall of Schlemm's canal and the ocular implant during delivery. The support provided by the dome-shaped wall may be represented by the force vectors shown in FIG. **19**. With reference to FIG. **19**, it will be appreciated that direction Dis generally parallel to the directions of first normal force vector FE and second normal force vector FF.

[0136] FIG. **20** is a perspective view showing an exemplary ocular implant **300** in accordance with this detailed description. Ocular implant **300** of FIG. **20** comprises a body **302** including a plurality of spines **304** and a plurality of frames **306**. The frames **306** of body **302** include a first frame **306**A, a second frame **306**B and a third frame **306**C.

[0137] First frame **306**A comprises a first strut **308**A and a second strut **308**B. First strut **308**A and second strut **308**B comprise a first landing surface **342**A and a second landing surface **342**B, respectively. First landing surface **342**A of first strut **308**A and second landing surface **342**B of second strut **308**B define a first footprint line **338**A. Second frame **306**B comprises a first strut **308**C and a second strut **308**D. First strut **308**C comprises a first landing surface **342**C and second strut **308**D comprises a second landing surface **342**D. First landing surface **342**C of first strut **308**C and second landing surface **342**D of fourth strut **308**D define a second footprint line **338**B. Third frame **306**C includes a first strut **308**E and a second strut **308**F. First strut **308**E and second strut **308**F have a first landing surface **342**E and a second landing surface **342**F, respectively. First landing surface **342**E of first strut **308**E and second landing surface **342**F of second strut **308**F define a third footprint line **338**C. In FIG. **20**, first footprint line **338**A, second footprint line **338**B, and third footprint line **338**C are shown intersecting at an apex **344** of a conical surface C.

[0138] Body **302** of ocular implant **300** has a distal end **326**, a proximal inlet portion **328** and an intermediate portion **330** extending between proximal inlet portion **328** and distal end **326**. Intermediate portion **330** of body **302** includes first frame **306**A, second frame **306**B, third frame **306**C and a plurality of spines **304**. The spines **304** of intermediate portion **330** include a first spine **304**A, a second spine **304**B and a third spine **304**C. In FIG. **20**, first frame **306**A can be seen extending between first spine **304**A and second spine **304**B. With reference to FIG. **20**, it will be appreciated that second frame **306**B extends between second spine **304**B and third spine **304**C. In the embodiment of FIG. **20**, third frame **306**C extends between third spine **304**C and distal end **326** of ocular implant **300** in FIG. **20**.

[0139] In the embodiment of FIG. **20**, each of first spine **304**A, second spine **304**B, and third spine **304**C are configured to preferentially bend in a direction that is at right angles to conical surface C defined by first footprint line **338**A, second footprint line **338**B, and third footprint line **338**C.

[0140] FIG. **21** is a stylized perspective view showing Schlemm's canal SC encircling an iris **30**. With reference to FIG. **21**, it will be appreciated that Schlemm's canal SC may overhang iris **30** slightly. Iris **30** defines a pupil **32**. Schlemm's canal SC forms a ring around iris **30** with pupil **32** disposed in the center of that ring. With reference to FIG. **21**, it will be appreciated that Schlemm's canal SC has a first major side **50**, a second major side **52**, a first minor side **54**, and a second minor side **56**. With reference to FIG. **21**, it will be appreciated that first major side **50** is further from pupil **32** than second major side **52**. In the exemplary embodiment of FIG. **21**, first major side **50** is an outer major side of Schlemm's canal SC and second major side **52** is an inner major side of Schlemm's canal SC.

[0141] For purposes of illustration, a window **70** is cut through first major side **50** of Schlemm's canal SC in FIG. **21**. Through window **70**, an ocular implant can be seen residing in a lumen defined by Schlemm's canal. The ocular implant shown in FIG. **21** is ocular implant **300** shown in the previous figure. Ocular implant **300** comprises a body **302** including a plurality of spines and a plurality of frames **306**. The frames **306** of body **302** include a first frame **306**A, a second frame **306**B and a third frame **306**C. First frame **306**A comprises a first landing surface **342**A and a second landing surface **342**B. First landing surface **342**A and second landing surface **342**B define a first footprint line **338**A. Second frame **306**B comprises a first landing surface **342**C and a second landing surface **342**D. First landing surface **342**C and second landing surface **342**D define a second footprint line **338**B. Third frame **306**C comprises a first landing surface **342**E and a second landing surface **342**F. First landing surface **342**E and second landing surface **342**F define a third footprint line **338**C. In the embodiment of FIG. **21**, first footprint line **338**A, second footprint line **338**B, and third footprint line **338**C intersect at an apex of a conical surface C. Due to page size constraints, conical surface C is truncated in FIG. **21**.

[0142] In the embodiment of FIG. **21**, the landing surfaces of each frame are configured to seat against the outer major side **50** of Schlemm's canal SC. In the eye, the outer major side of Schlemm's canal is backed by scleral tissue. Accordingly, in the exemplary embodiment of FIG. **21**, the landing surfaces of each frame will be seated against and supported by scleral tissue of the eye. Normal supporting forces will be applied to the landing surfaces of the struts by the scleral tissue. Applicant has created ocular implants designed to work in harmony with the dome shaped wall that defines the anterior chamber of the human eye. In some useful embodiments, the ocular implants are configured such that reaction forces applied to the ocular implant by scleral tissue while the ocular implant is being advanced into Schlemm's canal subject the ocular implant to pure bending with little or no twisting. The ocular implant may be configured such that a net twisting moment applied to each spine by the normal forces supporting the landing surfaces is substantially zero. The ocular implant may also be configured such that the normal forces supporting the landing surfaces primarily or exclusively act to bend each spine along the preferential bending plane thereof. In some useful embodiments, the preferential bending plane of each spine extends in a direction that is at right angles to a conical surface defined by the ocular implant.

[0143] While exemplary embodiments of the present invention have been shown and described, modifications may be made, and it is therefore intended in the appended claims to cover all such changes and modifications which fall within the true spirit and scope of the invention.

Claims

1. An ocular implant adapted to be disposed within Schlemm's canal of a human eye and configured to support Schlemm's canal in an open state, the ocular implant comprising: a body extending along a curved longitudinal central axis and configured to bend in a curvature plane, the body comprising a central channel, first and second frames and a spine interposed between the first and second frames, wherein the spine extends circumferentially from one side of the curvature plane and from an opposite side of the curvature plane, the body having dimensions adapted to be fit within Schlemm's canal; each frame comprising a first strut extending circumferentially on the one side of the curvature plane and a second strut extending circumferentially on the opposite side of the curvature plane, the first strut extending further out of the curvature plane than the corresponding second strut such that the first and second frames are not symmetric about the curvature plane.

2. The ocular implant of claim 1 wherein the body has a curved resting shape along the curvature plane.

3. The ocular implant of claim 1 wherein the spine comprises a first spine, the implant further comprising a third frame and a second spine interposed between the second and third frames, the third frame comprising a first strut on the one side of the curvature plane and a second strut on the opposite side of the curvature plane, the first strut of the third frame extending further out of the curvature plane than the corresponding second strut such that the third frame is not symmetric about the curvature plane.

4. The ocular implant of claim 3 wherein the first, second and third frames are substantially identical.

5. The ocular implant of claim 1 further comprising a first opening bordered by the first and second struts of the first frame and a second opening bordered by the first and second struts of the second frame, the first and second openings communicating with the central channel.

6. The ocular implant of claim 1 further comprising a channel opening bordered by circumferential extents of the spine and the first and second struts, the channel opening communicating with the

central channel.
7. The ocular implant of claim 1 wherein the curvature plane bisects the spine.
8. An ocular implant adapted to be disposed within Schlemm's canal of a human eye and configured to support Schlemm's canal in an open state, the ocular implant comprising: a body extending along a curved longitudinal central axis and configured to bend in a curvature plane, the body comprising a central channel, first and second frames and a spine interposed between the first and second frames, wherein the spine extends circumferentially from one side of the curvature plane and from an opposite side of the curvature plane, the body having dimensions adapted to be fit within Schlemm's canal; each frame comprising a first strut extending circumferentially on the one side of the curvature plane to a first landing surface and a second strut extending circumferentially on the opposite side of the curvature plane to a second landing surface, a footprint line drawn between the first landing surface and the second landing surface of each of the first frame and the second frame defining a roll angle with respect to the curvature plane of other than 90 degrees.
9. The ocular implant of claim 8 wherein the body has a curved resting shape along the curvature plane.
10. The ocular implant of claim 8 wherein the spine comprises a first spine, the implant further comprising a third frame and a second spine interposed between the second and third frames, the third frame comprising a first strut on the one side of the curvature plane extending circumferentially on the one side of the curvature plane to a first landing surface and a second strut on the opposite side extending circumferentially on the opposite side of the curvature plane to a second landing surface, a footprint line drawn between the first landing surface and the second landing surface of the third frame defining a roll angle with respect to the curvature plane of other than 90 degrees.
11. The ocular implant of claim IO wherein the first, second and third frames are substantially identical.
12. The ocular implant of claim 8 further comprising a first opening bordered by the first and second struts of the first frame and a second opening bordered by the first and second struts of the second frame, the first and second openings communicating with the central channel.
13. The ocular implant of claim 8 further comprising a channel opening bordered by circumferential extents of the spine and the landing surfaces of the first and second struts, the channel opening communicating with the central channel.
14. The ocular implant of claim 8 wherein the curvature plane bisects the spine.
